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Flowcell cartridge with floating bracket

Abstract

A cartridge for use with chemical or biological analysis systems, as well as methods of using the same, is provided. The cartridge may include a floating microfluidic plate that is held in the cartridge using one or more floating support brackets that incorporate gaskets that may seal against fluidic ports on the microfluidic plate. The floating support brackets may include indexing features that may align the microfluidic plate with the seals.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a divisional application under 35 U.S.C. § 120 of U.S. patent application Ser. No. 19/051,076, filed Feb. 11, 2025, which is itself a continuation of U.S. patent application Ser. No. 18/827,174, filed Sep. 6, 2024, which is itself a continuation of U.S. patent application Ser. No. 18/167,836, filed Feb. 11, 2023, which is a divisional of U.S. patent application Ser. No. 16/777,881, filed Jan. 30, 2020, and issued as U.S. Pat. No. 11,577,253 on Feb. 14, 2023, which is itself a divisional application under 35 U.S.C. § 120 of U.S. patent application Ser. No. 16/436,485, filed Jun. 10, 2019, and issued as U.S. Pat. No. 10,549,282 on Feb. 4, 2020, and which is itself a continuation of U.S. patent application Ser. No. 15/841,109, filed Dec. 13, 2017, which issued as U.S. Pat. No. 10,357,775 on Jul. 23, 2019, and which claims benefit of priority to United Kingdom (GB) application 1704769.7, filed Mar. 24, 2017, and also claims benefit of priority under 35 U.S.C. § 119(e) to U.S. Patent Application No. 62/441,927, filed Jan. 3, 2017, all of which are hereby incorporated by reference herein in their entireties.

BACKGROUND

(1) Sequencers, e.g., genome sequencers, such as DNA sequencers or RNA sequencers, and other biological or chemical analysis systems may sometimes utilize microfluidic flowcells, such as may be provided by way of a glass plate having microfluidic flow channels etched therein. Such flowcells may be made as a laminated stack of layers, with the flow channels etched in one or more of the layers. In most flowcells, access to the flow channels within the flowcell may be provided by way of openings that pass through one or both of the outermost layers to reach the flow channels within.

(2) Since it is difficult to decontaminate a flowcell after a sample has been flowed through it, it is common to replace the flowcell before analyzing a particular sample. As such, it is common for flowcells to be implemented using a cartridge-based approach to facilitate easy replacement of the flowcells.

SUMMARY

(3) Details of one or more implementations of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages will become apparent from the description, the drawings, and the claims. Note that the relative dimensions of the following figures may not be drawn to scale unless specifically indicated as being scaled drawings.

(4) In some implementations, an apparatus is provided that includes a frame, a microfluidic plate having one or more first fluidic ports in a first side, and a first support bracket that is attached to the frame such that the microfluidic plate is interposed between the first support bracket and the frame, the first support bracket floats relative to the microfluidic plate and the frame, the microfluidic plate and the frame float relative to one another, and a first side of the first support bracket faces towards the microfluidic plate. In such implementations, the first support bracket may include a first indexing feature that protrudes from the first side of the first support bracket and is proximate to a first edge of the microfluidic plate and may also include a second indexing feature that protrudes from the first side of the first support bracket and is proximate to a second edge of the microfluidic plate. The first support bracket may include a first gasket with at least one seal that is proud of the first side of the first support bracket and is positioned against the first side of the microfluidic plate, and the first indexing feature of the first support bracket and the second indexing feature of the first support bracket may contact the first edge and the second edge, respectively, of the microfluidic plate when the at least one seal of the first gasket is aligned with a corresponding at least one of the one or more first fluidic ports.

(5) In some such implementations, the microfluidic plate may have a second side opposite the first side, the frame may have a first overlapping portion that overlaps, when viewed along a direction perpendicular to a major surface of the microfluidic plate, a first portion of the microfluidic plate that includes the second edge, the first overlapping portion may be proximate to the second side of the microfluidic plate, the first overlapping portion may have a first clamp arm slot having a first slot width in a direction parallel to the second edge, the second side of the microfluidic plate may be visible, e.g., to the unaided eye, through the first clamp arm slot, the apparatus may be to, or configured to be, interfaced with a receiver of an analysis device, the receiver having a first clamp arm that is movable from an unclamped position in which the first clamp arm does not press on the second side of the microfluidic plate and does not engage with the first clamp arm slot to a clamped position in which the first clamp arm presses on the second side of the microfluidic plate and engages with the first clamp arm slot, and the first slot width may be larger than a width of the first clamp arm in a direction parallel to the second edge and located within the first clamp arm slot when the first clamp arm is in the clamped position.

(6) In some such implementations of the apparatus, the microfluidic plate may have a third edge opposite the first edge and a fourth edge opposite the second edge, the frame may have a second overlapping portion that overlaps, when viewed along the direction perpendicular to the major surface of the microfluidic plate, a second portion of the microfluidic plate that includes the fourth edge, the second overlapping portion may be proximate to the second side of the microfluidic plate, and the second overlapping portion may have a second clamp arm slot having a second slot width in a direction parallel to the fourth edge, the second side of the microfluidic plate may be visible through the second clamp arm slot, the receiver of the analysis device within which the apparatus is to be, or configured to be, interfaced may have a second clamp arm that is movable from an unclamped position in which the second clamp arm does not press on the second side of the microfluidic plate and does not engage with the second clamp arm slot to a clamped position in which the second clamp arm presses on the second side of the microfluidic plate and engages with the second clamp arm slot, and the second slot width may be larger than a width of the second clamp arm in a direction parallel to the fourth edge and located within the second clamp arm slot when the second clamp arm is in the clamped position.

(7) In some implementations of the apparatus, there may be two first fluidic ports in the microfluidic plate, and the first gasket may include two seals, each seal having a through-hole passing through the first support bracket and aligned with a different one of the first fluidic ports when the first indexing feature of the first support bracket and the second indexing feature of the first support bracket contact the first edge and the second edge, respectively, of the microfluidic plate.

(8) In some such implementations, the first gasket may include a support foot that is proud of the first side of the first support bracket and is positioned against the microfluidic plate, a first axis may be defined between center points of the two seals of the first gasket, the support foot of the first gasket may be offset by a first amount from the first axis along a second axis perpendicular to the first axis and parallel to the microfluidic plate, and the support foot of the first gasket may have an upper surface that contacts the microfluidic plate and is co-planar with upper surfaces of the two seals of the first gasket that are also in contact with the microfluidic plate. In some further such implementations of the apparatus, the support foot of the first gasket may not serve as a seal.

(9) In some implementations of the apparatus, the first gasket may be co-molded into the first support bracket.

(10) In some implementations of the apparatus, the first support bracket may have a second side that faces away from the first side of the first support bracket, and at least two first fluidic port indexing features may protrude from the second side of the first support bracket, each first fluidic port indexing feature to, or configured to, engage with a corresponding fluidic port indexing hole on a first fluidic port block of an analysis device to, or configured to, receive the apparatus.

(11) In some implementations of the apparatus, the frame may include two opposing first retaining clips with opposing surfaces that face one another, the first support bracket may be positioned in between the two opposing first retaining clips, the opposing surfaces of the first retaining clips may be spaced apart by a first distance, and the portion of the first support bracket between the opposing surfaces of the first retaining clips may have a first width in a direction spanning between the opposing surfaces of the first retaining clips that is less than the first distance.

(12) In some implementations of the apparatus, the first support bracket may include a third indexing feature that protrudes from the first side of the first support bracket and is proximate to a third edge of the microfluidic plate opposite the first edge of the microfluidic plate, and the microfluidic plate may be interposed between the first indexing feature of the first support bracket and the third indexing feature of the first support bracket.

(13) In some implementations of the apparatus, the microfluidic plate may be rectangular and the first edge of the microfluidic plate may be orthogonal to the second edge of the microfluidic plate and the second edge of the microfluidic plate may be orthogonal to the third edge of the microfluidic plate.

(14) In some implementations of the apparatus, the frame may have a substantially rectangular opening, the microfluidic plate may sit within the substantially rectangular opening, the substantially rectangular opening may have opposing side walls that face towards one another, and the first indexing feature of the first support bracket may be interposed between one of the opposing side walls of the substantially rectangular opening and the first edge of the microfluidic plate and the third indexing feature of the first support bracket may be interposed between the other opposing side wall of the opposing side walls of the substantially rectangular opening and the third edge of the microfluidic plate.

(15) In some implementations of the apparatus, the substantially rectangular opening may have an opening width in a direction parallel to the second edge, a first indexing feature width may exist between furthest-apart portions of the surfaces of the first indexing feature of the first support bracket and the third indexing feature of the first support bracket that face the opposing side walls of the substantially rectangular opening, and the opening width minus the first indexing feature width may be less than the first distance minus the first width.

(16) In some implementations, the microfluidic plate may further include one or more second fluidic ports on the first side and the apparatus may further include a second support bracket that is attached to the frame such that the microfluidic plate is interposed between the second support bracket and the frame, the second support bracket floats relative to the microfluidic plate and the frame, the microfluidic plate and the frame float relative to one another, and a first side of the second support bracket faces towards the microfluidic plate. In such implementations, the second support bracket may include a first indexing feature that protrudes from the first side of the second support bracket and is proximate to the first edge of the microfluidic plate, the second support bracket may include a second indexing feature that protrudes from the first side of the second support bracket and is proximate to a fourth edge of the microfluidic plate opposite the second edge of the microfluidic plate, the microfluidic plate may be interposed between the second indexing feature of the first support bracket and the second indexing feature of the second support bracket, the second support bracket may include a second gasket with at least one seal that is proud of the first side of the second support bracket and is positioned against the microfluidic plate, and the first indexing feature of the second support bracket and the second indexing feature of the second support bracket may contact the first edge and the fourth edge, respectively, of the microfluidic plate when the at least one seal of the second gasket is aligned with a corresponding at least one of the one or more second fluidic ports.

(17) In some such implementations, the frame may include two opposing second retaining clips with opposing surfaces that face one another, the second support bracket may be positioned in between the two opposing second retaining clips, the opposing surfaces of the second retaining

clips may be spaced apart by a second distance, and the portion of the second support bracket between the opposing surfaces of the second retaining clips may have a second width in a direction spanning between the opposing surfaces of the second retaining clips that is less than the second distance.

(18) In some further such implementations, the second support bracket may include a third indexing feature that protrudes from the first side of the second support bracket and is proximate to the third edge of the microfluidic plate, and the microfluidic plate may be interposed between the first indexing feature of the second support bracket and the third indexing feature of the second support bracket.

(19) In some additional such implementations, the frame may have a substantially rectangular opening, the microfluidic plate may have a third edge opposite the first edge, the microfluidic plate may sit within the substantially rectangular opening, the substantially rectangular opening may have opposing side walls that face towards one another and that define an opening width in a direction parallel to the second edge, the first indexing feature of the second support bracket may be interposed between one of the opposing side walls of the substantially rectangular opening and the first edge of the microfluidic plate and the third indexing feature of the second support bracket may be interposed between the other opposing side wall of the opposing side walls of the substantially rectangular opening and the third edge of the microfluidic plate, the microfluidic plate may have a plate width in a direction spanning between the first indexing feature of the second support bracket and the third indexing feature of the second support bracket, a second indexing feature width may exist between furthest-apart portions of the surfaces of the first indexing feature of the second support bracket and the third indexing feature of the second support bracket that face the opposing side walls of the substantially rectangular opening, and the opening width minus the second indexing feature width may be less than the second distance minus the second width.

(20) In some implementations, there may be two second fluidic ports in the microfluidic plate, and the second gasket may include two seals, each seal having a through-hole passing through the second support bracket and aligned with a different one of the second fluidic ports when the first indexing feature of the second support bracket and the second indexing feature of the second support bracket contact the first edge and the fourth edge, respectively, of the microfluidic plate.

(21) In some implementations, the second gasket may include a support foot that is proud of the first side of the second support bracket and is positioned against the microfluidic plate, a third axis may be defined between center points of the two seals of the second gasket, the support foot of the second gasket may be offset by a second amount from the third axis along a fourth axis perpendicular to the third axis and parallel to the microfluidic plate, and the support foot of the second gasket may have an upper surface that contacts the microfluidic plate and may be co-planar with upper surfaces of the two seals of the second gasket that are also in contact with the microfluidic plate. In some such implementations, the support foot of the second gasket may not serve as a seal. In some alternative or additional such implementations, the second gasket may be co-molded into the second support bracket.

(22) In some implementations, the second support bracket may have a second side that faces away from the first side of the second support bracket, and at least two second fluidic port indexing features may protrude from the second side of the first support bracket, each first fluidic port indexing feature to, or configured to, engage with a corresponding fluidic port indexing hole on a first fluidic port block of an analysis device to, or configured to, receive the apparatus.

(23) These and other implementations are described in further detail with reference to the Figures and the detailed description below. Other features, aspects, and advantages will become apparent from the description, the drawings, and the claims. Note that the relative dimensions of the following figures may not be drawn to scale.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The various implementations disclosed herein are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings, in which like reference numerals refer to similar elements.
- (2) FIG. 1 depicts an exploded isometric view of an example flowcell cartridge.
- (3) FIG. 2 depicts an exploded underside isometric view of the example flowcell cartridge of FIG. 1.
- (4) FIG. 3 depicts a front isometric view of the example flowcell cartridge of FIG. 1 in an unexploded state.
- (5) FIG. 4 depicts a rear isometric view of the example flowcell cartridge of FIG. 1 in an unexploded state.
- (6) FIGS. 5 and 6 are diagrams illustrating how a seal can roll when the surfaces between which the seal is interposed are translated laterally.
- (7) FIGS. 7 and 8 are diagrams illustrating how a gasket with a support foot can prevent the rolling behavior illustrated in FIGS. 5 and 6.
- (8) FIG. 9 depicts an isometric view of a floating support bracket of the example flowcell cartridge of FIG. 1.
- (9) FIG. 10 depicts an underside isometric view of the floating support bracket of the example flowcell cartridge of FIG. 1.
- (10) FIG. 11 depicts an isometric view of an example receiver for the example flowcell cartridge of FIG. 1.
- (11) FIG. 12 depicts an exploded isometric view of the example receiver of FIG. 11 and the example flowcell cartridge of FIG. 1.
- (12) FIG. 13 depicts a plan view of the example flowcell cartridge of FIG. 1.
- (13) FIGS. 14 through 17 depict various stages of component alignment that may occur during clamping of an example flowcell cartridge.
- (14) FIGS. 1 through 4 and 9 through 13 are drawn to scale within each Figure, although the scale of the depicted embodiments may vary from Figure to Figure.

DETAILED DESCRIPTION

- (15) The present inventors have conceived of new designs for a flowcell cartridge, such as may be used in chemical and biological analysis systems that utilize microfluidic flow structures contained within a glass plate structure. These concepts are discussed herein with respect to the following Figures, although it will be appreciated that these concepts may be implemented in cartridge designs other than the specific example shown, and that such other implementations would still potentially fall within the scope of the claims.
- (16) FIG. 1 depicts an exploded isometric view of an example flowcell cartridge. In FIG. 1, the flowcell cartridge 100 has a frame 102 that may, for example, be made of molded plastic or other, durable material. The frame may provide a support structure for supporting a glass plate (or a plate of other material, e.g., acrylic or other plastic), such as glass plate 114 that contains microfluidic flow structures; this plate may also be referred to herein as a microfluidic plate. In this example, the glass plate, which has a first edge 122, a second edge 124, a third edge 126, and a fourth edge 128, includes four sets of multiple, parallel microfluidic flow channels that extend along directions parallel to the long axis of the glass plate, e.g., along axes that are parallel to the first edge 122 and/or the third edge 126. To the extent applicable, the terms “first,” “second,” “third,” etc. (or other ordinal indicators) herein are merely employed to show the respective objects described by these terms as separate entities and are not meant to connote a sense of chronological order, unless stated explicitly otherwise herein. The first edge 122 and the third edge 126 may be generally

orthogonal to the second edge **124** and the fourth edge **128** in some implementations, but may be other orientations in other implementations. As can be seen in FIG. 2, which depicts an exploded underside isometric view of the example flowcell cartridge of FIG. 1, each set of microfluidic flow structures may terminate in one or more first fluidic ports **118** and one or more second fluidic ports **120**. The first and second fluidic ports **118** and **120** may be located in a first side **116** of the glass plate **114**, although other implementations may only include the first fluidic ports **118** or the second fluidic ports **120** on the first side **116**. The frame **102** may have a substantially rectangular opening (or opening of another shape) **104** that is sized to receive the glass plate **114**; the rectangular opening **104** may include opposing side walls **106** that are in close proximity to the first edge **122** and the third edge **126** of the glass plate **114** when the cartridge is fully assembled. As used herein, the term “substantially rectangular” is used to refer to an opening that has an overall rectangular shape, although there may be various features or discontinuities in the overall shape, such as the semi-circular notches along one side wall of the depicted rectangular opening, or the clamp arm slots along the short edges of the rectangular opening **104**. The opposing side walls **106** may be spaced apart by an opening width **195** to allow the first support bracket **132** and the second support bracket **160**, and thus the glass plate **114**, to float within the rectangular opening **104** for at least some range of movement, e.g., about 1 mm to about 2 mm or less.

(17) The glass plate **114** may be held in place in the cartridge **100** through the use of one or more support brackets, such as a first support bracket **132** and a second support bracket **160**. In this discussion, only the features of the first support bracket **132** are discussed in detail, although it is readily apparent from the Figures that the second support bracket **160**, which may or may not be identical to the first support bracket **132**, is at least structurally similar to the first support bracket **132** and may operate in a similar manner.

(18) The first support bracket **132** may have a first side **134** (see FIG. 1) and a second side **136** (see FIG. 2). The first side **134** may face towards the glass plate **114** and may have a first indexing feature **138**, e.g., a molded pin or post, that extends away from the first side **134** and that is at least long enough that the side of the first indexing feature **138** that faces towards the glass plate **114** may contact the glass plate **114** when the cartridge is fully assembled. The first indexing feature **138** may be positioned on the first support bracket **132** such that the first indexing feature **138** is proximate to, or contacting, the first edge **122** of the glass plate **114** when the cartridge is fully assembled. The first support bracket **132** may also have one or more second indexing features **140** (an additional second indexing feature **140'** is also shown in FIG. 1) that may be similar to the first indexing feature **138** except that each second indexing feature **140** may be positioned on the first support bracket **132** such that the second indexing feature **140** is proximate to, or physically contacts, the second edge **124** of the glass plate **114**. The first support bracket **132** may also include a third indexing feature **142**, which may be positioned on an opposite end of the first support bracket **132** from the first indexing feature **138**. The first indexing feature **138** and the third indexing feature **142**, if used, may be separated from one another by a first float gap **156**, which may be sized to be slightly larger than the plate width **130** so as to allow the glass plate **114** to “float” within the confines of the first indexing feature **138** and the third indexing feature **142**. The furthest-apart surfaces of the first indexing feature **138** and the third indexing feature **142** may similarly define a first indexing feature width **157**. The opening width **195** may be wider than the first indexing feature width **157** so that the first support bracket **132** may float laterally between the opposing side walls **106** of the rectangular opening **104**.

(19) The first support bracket may also include one or more first gaskets **144**, which may include one or more seals **146** (each first gasket **144**, in this example, includes two seals **146**, each positioned so as to interface with a different first fluidic port **118**). The first gaskets **144** may, for example, be insertable into the first support bracket **132** or may, in some implementations, be co-molded with the first support bracket **132** (in the latter case, the first gaskets **144** and the first support bracket **132** may, in effect, be treated as a single component). The seals may be proud of

the first side **134** and, optionally, the second side **136** of the first support bracket so that they may compress against the glass plate **114** and, as discussed later herein, a fluidic port block, respectively. In some implementations, the seal may not be proud of the second side **136** of the first support bracket, e.g., if the fluidic port block that faces the second side **136** when the cartridge is installed in an analysis device has a raised boss that may engage with the seal.

(20) The first gasket **144** may also include a support foot **148**, which may be provided to prevent or mitigate “rolling” of the first gasket **144** about an axis passing through the centers of the seals **146** when the first support bracket **132** is translated in a direction parallel to the major surface of the glass plate **114** while the seals **146** are in contact with the glass plate **114**. To this end, the support foot **148** may be offset from a first axis **150** spanning between the centers of the seals **146** of the first gasket **144** along a second axis **152** perpendicular to the first axis **150** by some amount so as to provide a moment arm to resist such rolling behavior. The support foot **148** and the seals **146** may all be designed to have contact surfaces that contact the glass plate **114** in concert when the glass plate **114** is brought into contact with the first gasket **144**. These contact surfaces may all be parallel to one another to ensure that when the contact surface of the support foot **148** is in contact with the glass plate **114**, the contact surface(s) of the seal(s) **146** are also in good, i.e., not having any misalignment gaps, contact with the glass plate **114**. In the example cartridge shown, each support bracket includes two first gaskets, although they may be referred to as second gaskets, third gaskets, etc., in the interests of reducing confusion, if needed. It is also be understood that the support foot **148**, while appearing similar to the seals **146**, may actually not provide any “sealing” characteristics at all—it may be present solely for the purposes of preventing or mitigating “rolling.”

(21) FIGS. **5** and **6** are diagrams illustrating how a seal can roll when the surfaces between which the seal is interposed are translated laterally. In FIG. **5**, a glass plate **514** is offset from a fluidic port block **564**, and a support bracket **532** with a gasket **544** is interposed between them. The gasket **544** has a seal **546** that is aligned with a fluidic port **518'** in the fluidic port block **564**, but that is misaligned somewhat with a fluidic port **518** in the glass plate **514**. As can be seen in FIG. **6**, when the glass plate **514** is slid sideways so that the fluidic port **518** is aligned with the seal **546**, friction between the seal **546** and the glass plate **514**/fluidic port block **564** may cause the seal **546** to not slide a commensurate distance—as a result, the gasket **544** and the support bracket **532** may tilt or roll slightly, resulting in gaps **594** appearing between the seal **546** and the glass plate **514**/fluidic port block **564**. This is, of course, undesirable, as it causes leakage.

(22) FIGS. **7** and **8** are diagrams illustrating how a gasket with a support foot can prevent the rolling behavior illustrated in FIGS. **5** and **6**. As can be seen, the gasket **544** has been extended to the right and a support foot **748** has been added to the gasket **544**. When the glass plate **514** is slid to the left, as in FIG. **6**, the support foot **748** introduces a counter-moment to any potential rolling moment caused by friction between the seal **546** and the glass plate **514**/fluidic port block **564**. This prevents the formation of the gaps **594** and keeps the seal **546** in good contact with the surfaces it seals.

(23) The first support bracket **132** may snap into two opposing first retaining clips **108** (only one is visible in FIG. **2**, as the other is obscured by other features of the frame **102**—however, there are corresponding second retaining clips visible on the opposite end of the frame **102** that are configured similarly but at a different location). The first retaining clips **108** may have opposing surfaces **110** that are separated from one another by a first distance **112**. The first distance may be greater than a first width **158** of the first support bracket **132**, thereby allowing the first support bracket **132** to float laterally by a small amount when snapped into the first retaining clips **108**. In some implementations, the amount of float between the first support bracket **132** and the opposing side walls **106**, i.e., the opening width **195** minus the first indexing feature width **157**, may be smaller than the amount of float between the first support bracket **132** and the retaining clips **108**, i.e., the first distance **112** minus the first width **158**. Similar relationships may exist for the second support bracket **160**.

(24) FIG. 3 depicts a front isometric view of the example flowcell cartridge of FIG. 1 in an unexploded/assembled state. FIG. 4 depicts a rear isometric view of the example flowcell cartridge of FIG. 1 in an unexploded/assembled state. As can be seen, the glass plate **114** is held in place within the frame **102** by the first support bracket **132** and the second support bracket **160**, which, in turn, are held in place by the first retaining clips **108** and second retaining clips, respectively. The frame may have a first overlapping portion **196** and a second overlapping portion **196'** (see FIG. 2) that overlap with a corresponding first portion **197** and second portion **197'** (see FIG. 1) of the glass plate **114**. The first portion **197** may include the second edge **124**, and the second portion **197'** may include the fourth edge **128**. The overlapping portions **196/196'** may prevent the glass plate **114** from falling out of the front of the frame **102**, e.g., the glass plate **114** may be sandwiched between the overlapping portions **196/196'** and the first/second support brackets **132/160**. The glass plate **114** may still, however, be free to float within the frame to some degree.

(25) FIG. 9 depicts an isometric view of the first support bracket **132** of the example flowcell cartridge **100** of FIG. 1. FIG. 10 depicts an underside isometric view of the first support bracket **132** of the example flowcell cartridge **100** of FIG. 1. In addition to the first indexing feature **138**, the second indexing feature(s) **140**, and possibly the third indexing feature **142**, the first support bracket **132** may also include first fluidic port indexing features **154** on the second side **136** of the first support bracket **132** (the second support bracket **160** may have corresponding second fluidic port indexing features as well). As can be seen, the first support bracket has portions that extend beyond the first width **158**, e.g., the small “teeth” that are located at the four outermost corners of the first support bracket **132**. These teeth may engage with the first retaining clips **108** and may allow the first support bracket **132** to also float along an axis parallel to the first edge **122** by some limited amount.

(26) In this example cartridge, the glass plate **114** may float with respect to the support brackets **132** and **160**, and the support brackets **132** and **160**, in turn, may float with respect to the frame **102**. Thus, there are two tiers of floating components in the example cartridge. The combination of these different tiers of floating components, as well as the various indexing features provided, allow for the glass plate **114** and the seals **146** to be properly aligned with each other and with ports on floating manifold blocks located on equipment that receives the cartridge **100**.

(27) FIG. 11 depicts an isometric view of an example receiver for the example flowcell cartridge of FIG. 1. As seen in FIG. 11, a receiver **162** may be provided; the receiver may be a subcomponent of a larger analysis device that utilizes the cartridge **100**. The receiver **162** may include a chuck **176**, against which the glass plate **114** may be drawn, e.g., by a vacuum, during analysis operations. The receiver **162**, in this example, may include a pair of first fluidic port blocks **164** and an opposing pair of second fluidic port blocks **166**. The first fluidic port blocks **164** and the second fluidic port blocks **166** may be configured to float slightly in directions at least parallel to the upper surface of the chuck **176** (and possibly also in directions perpendicular to the upper surface of the chuck **176**). The ends of the receiver **162** may include, for example, a clamping mechanism that may serve to clamp the glass plate **114** against the chuck **176**. Such clamping mechanisms may, for example, have clamp arms **172** that may rotate downwards and contact the upper surface of the glass plate **114** of the cartridge **100** when the cartridge **100** is installed. The receiver **162** may also include indexing features that are located so as to engage with the support brackets and glass plate **114** of the cartridge **100** when the cartridge **100** is installed. For example, lateral indexing pins **168** may be placed such that the glass plate **114** contacts the lateral indexing pins **168** when the glass plate **114** is translated laterally along the short axis of the chuck **176**, and longitudinal indexing pins **170** may be positioned so as to contact the support brackets of the cartridge **100** when, for example, one of the longitudinal indexing pins **170** is moved towards the other longitudinal indexing pins **170**. In this example, the longitudinal indexing pin **170** on the left is fixed in space relative to the receiver **162**, whereas the other longitudinal indexing pin **170** is configured to slide along an axis parallel to the long axis of the chuck **176**. The sliding longitudinal indexing pin **170** may be sprung so as to be

biased towards the other longitudinal indexing pin **170**. The interaction of the various indexing features is explained in more detail below, with respect to FIG. **12**.

(28) FIG. **12** depicts an exploded isometric view of the example receiver of FIG. **11** and the example flowcell cartridge of FIG. **1**. In this example, the cartridge **100** has been shown in an exploded view, although the various components that form the cartridge would be fully assembled, per FIG. **3**, prior to the cartridge **100** being placed in the receiver **162**.

(29) When the cartridge **100** is laid on top of the receiver **162**, the clamp arms **172** may rotate downward and engage with the top side of the glass plate **114**. The clamp arms **172** may also, as they pivot, translate along their rotational axes towards the lateral indexing pins **168** such that the sides of the clamp arms **172** engage with the sides of the rectangular notches or clamp arm slots **198**, thereby causing the entire frame **102** to translate along the same axis as well. For example, the clamp arm slots **198** may be sized, e.g., with clamp arm widths **173** in a direction parallel to the second edge **124** that are less than the widths of the clamp arm slots **198** in the same direction, to allow the clamp arms **172** to swing through the clamp arm slots **198** freely and, during lateral translation of the clamp arms **172**, press against the sides of the clamp arm slots **198** facing away from the lateral indexing pins **168**, thereby pushing the frame **102** towards the lateral indexing pins **168**. During this lateral sliding motion, the frame **102** will (if not already in such a state) come into contact with the first indexing feature **138** on the first support bracket **132** (and a corresponding first indexing feature on the second support bracket **160**) at indexing feature contact points **182** located along one of the opposing side walls **106**. As the frame **102** continues to be translated towards the lateral indexing pins **168**, the glass plate **114** will eventually come into contact with both the lateral indexing pins **168** and the first indexing features **138** (see lateral indexing pin contact points **184** and the indexing feature contact points **182** along the first edge **122** of the glass plate **114**). Eventually, the first indexing features **138** will be sandwiched between the frame **102** and the glass plate **114** (which is pressed against the lateral indexing pins **168**), thereby locating the first support bracket **132** and the second support bracket **160** firmly in space in the lateral direction, i.e., perpendicular to the long axis of the chuck **176**. This aligns the seals on the first support bracket **132** and the second support bracket **160** with the corresponding first fluidic ports **118** and the corresponding second fluidic ports **120**, respectively, on the glass plate **114**.

(30) Subsequent to, after, or in concert with the translation of the frame **102** towards the lateral indexing pins **168**, the longitudinal indexing pins **170** may be caused to move towards one another (one or both may move), thereby contacting the facing edges of the first support bracket **132** and the second support bracket **160** and pushing the first support bracket **132** and the second support bracket **160** towards one another. As the first support bracket **132** and the second support bracket **160** move towards one another, the glass plate **114** may come into contact with the second indexing features **140** (and **140'**, if present) on the first support bracket **132** and the second support bracket **160**. The first support bracket **132** and the second support bracket **160** may thus become aligned with the glass plate **114** and, consequently, the first fluidic ports **118** and the second fluidic ports **120**.

(31) After or during such plate alignment, the fluidic port blocks **164**, **166** may be raised so that the first fluidic port indexing features **154** (and corresponding second fluidic port indexing features on the second support bracket **160**) may be inserted into corresponding alignment holes **188** on the first fluidic port block **164** and the second fluidic port block **166**. As the fluidic port block rises, the first fluidic port indexing features **154** and the second fluidic port indexing features may engage with the corresponding alignment holes **188** and force the first fluidic port blocks **164** and the second fluidic port blocks **166** into alignment with the first support bracket **132** and the second support bracket **160**, respectively. This, in turn, ensures that the corresponding seals **146** on the respective support brackets **132**, **160** line up with the fluidic ports on the first fluidic port blocks **164** and the second fluidic port blocks **166**, respectively.

(32) Thus, the cartridge **100** may have multiple levels of floating components that engage with

different sets of indexing features/pins in the cartridge **100** and located on the receiver **162** and are moved into precisely aligned positions that cause the fluidic ports, seals, and port block ports to line up, e.g., such that the centerlines of the fluidic ports, seals, and port block ports are, in some implementations, within less than about 0.05 mm of one another, thereby ensuring a high-quality liquid-tight seal. At the same time, some implementations of the cartridge may feature additional features in the floating brackets, e.g., support feet, that may prevent rolling behavior of the seal, thereby ensuring the integrity of any sealed connections. Some of the floating components, e.g., the support brackets, may also act to retain other floating components, e.g., the glass plate, in a manner that prevents stressing the glass plate due to thermal expansion mismatches between the glass plate and the cartridge frame, minor flexure of the cartridge frame, and so forth.

(33) The floating behavior of the various components in the cartridge **100** may be better understood with reference to FIG. **13**, which depicts a plan view of the example flowcell cartridge of FIG. **1**. For reference purposes, the lateral indexing pins **168** are shown as dotted circles and the outlines of the clamp arms **172** are shown as dotted, rounded rectangles, but the remainder of the components shown are part of the cartridge **100**. The clamp arms **172** are shown in both an “engaged” position (black line font) in which they are engaged with and pressed against the sides of the clamp arm slots **198** (see FIG. **2**) and a non-engaged position (grey line font), which may be their position prior to translating laterally. The glass plate **114** may be able to move laterally by an amount relative to the frame **102** that is limited by the first and second indexing features **138** and **142**, respectively **11**. The first and second support brackets may be able to move laterally (as well as longitudinally) by a lesser amount, as is shown by the bracket float envelopes **180**. For example, the first and second support brackets may be able to float laterally by a distance of X , which may be the opening width **195** minus the first indexing feature width **157**, relative to the frame, and the glass plate **114** may be able to float laterally by a distance of Y , which may be the first float gap **156** minus the plate width **130**, relative to the first and second support brackets **132** and **160**. In some such implementations, Y may be less than X —however, the glass plate **114** may still float by a larger amount relative to the frame **102** than the first and second support brackets **132** and **160** since the glass plate **114** has a total overall float relative to the frame **102** of $X+Y$. This may allow for considerable adjustment in the positioning of the glass plate.

(34) An example alignment sequence is reviewed in FIGS. **14** through **17**, which depict various stages of component alignment that may occur during clamping of an example flowcell cartridge. In FIG. **14**, the frame **1402** (shown in solid lines) of a flowcell cartridge is lowered onto a receiver with two floating fluidic port blocks **1464** (shown in dashed lines). As can be seen, the fluidic port blocks **1464** are slightly askew due to the fact that both are “floating.” Also visible in FIG. **14** is the outline of a support bracket **1432** (dotted lines) and a glass plate **1414** (dash-dot-dash lines). There are four instances of fluidic ports **1418** across the glass plate **1414**. As can be seen, at each fluidic port **1418**, there are corresponding features belonging to the support bracket (dotted circles) and fluidic port blocks (dashed lines). These correspond, for example, to the holes in the seals **146** and to the ports in the fluidic port blocks **1464**. As is evident, there is some alignment between these three separate fluidic flow features at each location, but the alignment is far from ideal, resulting in differently-configured apertures at each location which may cause imbalances in fluid flow.

(35) In FIG. **15**, the support bracket **1432** has been fully engaged with the fluidic port blocks **1464** so that fluidic port indexing features **1454** (see FIG. **14**) are fully inserted into alignment holes **1488** (also see FIG. **14**). The alignment holes **1488**, for example, may be countersunk and the fluidic port indexing features **1454** may have conical or rounded tips so that they may engage with one another even if somewhat misaligned; as the fluidic port indexing features **1454** are more fully engaged with the alignment holes **1488**, the countersink portion may narrow and force the fluidic port indexing features **1454** to move towards the center of the alignment holes **1488**. As can be seen, one of the alignment holes **1488** for a given fluidic port block **1464** may be circular, thereby providing both X and Y location constraints, whereas the other may be obround to provide a single

degree of constraint, e.g., along only the Y axis, as this may be all that is needed in one implementation to prevent rotation about the other alignment hole **1488**. It is to be recognized that the alignment holes **1488** and the fluidic port indexing features **1454** may also be swapped, i.e., the alignment holes **1488** may be located on the support bracket **1432**, and the fluidic port indexing features **1454** may be located on the fluidic port block **1464**.

(36) Returning to FIG. **15**, the interfacing of the cartridge with the fluidic support blocks **1464** causes the fluidic port blocks **1464** to come into alignment with each other as well as with the support bracket **1432**. Consequently, the ports on the fluidic port blocks **1464** are now precisely aligned with the holes, e.g., the seals, on the support bracket **1432**. However, the holes/seals on the support bracket **1432** are not yet aligned with the fluidic ports **1418** on the glass plate.

(37) In FIG. **16**, the glass plate **1414** has been moved upwards to contact second indexing features **1440** on the support bracket **1432**; this contact and the upward movement of the glass plate **1414** causes the support bracket **1432** to move upwards until it contacts longitudinal indexing pin **1470**, thus firmly locking the support bracket **1432** in place in the vertical direction (with respect to the Figure orientation; in reality, this is more accurately called the longitudinal direction)—this aligns the fluidic ports **1418** in the glass plate **1414** with the corresponding holes/seals in the support bracket **1432** in the vertical direction.

(38) Finally, in FIG. **17**, the frame **1402** may be pushed towards the lateral indexing pin **1468**. This causes the inside edge of the frame **1402** to contact first indexing feature **1438**, which causes the support bracket **1432**, in turn, to move towards the lateral indexing pin **1468** until the first indexing feature **1438** also contacts the glass plate **1414** and pushes the opposite side of the glass plate **1414** into contact with the lateral indexing pin **1468**. As can be seen, the first fluidic ports **1418** and the respective seal holes and fluidic port block holes are completely aligned, thereby ensuring a consistently-sized flow aperture and proper seal alignment.

(39) The term “about” used throughout this disclosure, including the claims, is used to describe and account for small fluctuations, such as due to variations in processing. For example, unless otherwise specified herein in a particular context, they can refer to less than or equal to $\pm 5\%$, of the specified value or value equivalent to the specified relationship, such as less than or equal to $\pm 2\%$, such as less than or equal to $\pm 1\%$, such as less than or equal to $\pm 0.5\%$, such as less than or equal to $\pm 0.2\%$, such as less than or equal to $\pm 0.1\%$, such as less than or equal to $\pm 0.05\%$.

(40) As noted earlier, any use of ordinal indicators, e.g., (a), (b), (c) . . . or the like, in this disclosure and claims is to be understood as not conveying any particular order or sequence, except to the extent that such an order or sequence is explicitly indicated. For example, if there are three steps labeled (i), (ii), and (iii), it is to be understood that these steps may be performed in any order (or even concurrently, if not otherwise contraindicated) unless indicated otherwise. For example, if step (ii) involves the handling of an element that is created in step (i), then step (ii) may be viewed as happening at some point after step (i). Similarly, if step (i) involves the handling of an element that is created in step (ii), the reverse is to be understood.

(41) It is also to be understood that the use of “to,” e.g., “the apparatus is to be interfaced with a receiver of an analysis device,” may be replaceable with language such as “configured to,” e.g., “the apparatus is configured to be interfaced with a receiver of an analysis device”, or the like.

(42) It should be appreciated that all combinations of the foregoing concepts (provided such concepts are not mutually inconsistent) are contemplated as being part of the inventive subject matter disclosed herein. In particular, all combinations of claimed subject matter appearing at the end of this disclosure are contemplated as being part of the inventive subject matter disclosed herein. For the sake of brevity, many of those permutations and combinations will not be discussed and/or illustrated separately herein.

Claims

1. A chemical or biological analysis system comprising: a microfluidic cartridge that includes: a frame; a microfluidic plate positioned within the frame, wherein the microfluidic plate floats relative to the frame, the microfluidic plate comprising a plurality of first fluidic ports; and a support positioned within the frame, wherein the support floats relative to the microfluidic plate and the frame, the support comprising: a gasket having a plurality of seals, and a plurality of alignment holes; and a sequencer with a receiver that includes: a chuck; a plurality of analysis device ports; a first plurality of pins configured to extend into the plurality of alignment holes and align the plurality of analysis device ports and the plurality of seals; and a second plurality of pins configured to engage the microfluidic plate such that the plurality of first fluidic ports and the plurality of seals are aligned, wherein each seal of the plurality of seals has a through-hole passing through the seal to fluidically connect the plurality of first fluidic ports to the analysis device ports.
 2. The chemical or biological analysis system of claim 1, wherein a plurality of teeth extends from the support, the plurality of teeth configured to engage with the frame and partially constrain movement of the support relative to the frame.
 3. The chemical or biological analysis system of claim 1, wherein the plurality of analysis device ports is located in one or more fluidic port blocks.
 4. The chemical or biological analysis system of claim 1, wherein the plurality of alignment holes comprises an alignment hole having a countersink portion that narrows to direct a corresponding pin of the first plurality of pins toward a center of the alignment hole.
 5. The chemical or biological analysis system of claim 1, wherein the plurality of alignment holes comprises a circular alignment hole to constrain movement of the support along perpendicular axes of the microfluidic plate and an obround alignment hole to constrain movement of the support along at least one of the perpendicular axes.
 6. The chemical or biological analysis system of claim 1, wherein: the frame has a substantially rectangular opening, the microfluidic plate sits within the substantially rectangular opening, the substantially rectangular opening has opposing side walls that face towards one another, and a plurality of apertures is located in one of the opposing side walls.
 7. The chemical or biological analysis system of claim 1, wherein the microfluidic cartridge further includes a second support positioned within the frame, wherein the second support floats relative to the microfluidic plate and the frame, the second support comprising: a second gasket having a plurality of seals, and a second plurality of alignment holes.
 8. The chemical or biological analysis system of claim 7, wherein the receiver further includes: a second plurality of analysis device ports; and a third plurality of pins corresponding to the second plurality of alignment holes, wherein the third plurality of pins extends into the second plurality of alignment holes and align the second plurality of analysis device ports and the plurality of seals of the second gasket are aligned.
 9. The chemical or biological analysis system of claim 8, wherein the second support further comprises a projection extending from an edge of the second support, the projection configured to engage with the frame and limit movement of the second support relative to the frame.
 10. The chemical or biological analysis system of claim 1, wherein the receiver further includes a clamp comprising a plurality of clamp arms configured to press the microfluidic cartridge against the chuck.
 11. The chemical or biological analysis system of claim 3, wherein the one or more fluidic port blocks float in a direction at least parallel to an upper surface of the chuck of the sequencer.
 12. The chemical or biological analysis system of claim 6, wherein the frame comprises a portion that overlaps a portion of the microfluidic plate.
 13. The chemical or biological analysis system of claim 6, wherein the plurality of apertures comprises two notches.
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